

Redes de Computadores

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Comunicação

Distância

Custo

Comodidade

Segurança

Tempo



Mensageiros

Telégrafo hidráulico

Fonte: <https://www.youtube.com/watch?v=Az28Hsne-nI>



Torre de Hércules

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Tambores Africanos

A man in a light blue shirt and dark trousers stands on the deck of a ship, viewed from behind. He is holding two signal flags high in the air with both arms. The flag in his right hand is red, and the flag in his left hand is yellow. The background shows a clear blue sky and the ocean. Other ships are visible in the distance, including one with a prominent mast and radar equipment. The text "Sinalização por bandeiras" is overlaid in the bottom right corner.

**Sinalização
por bandeiras**



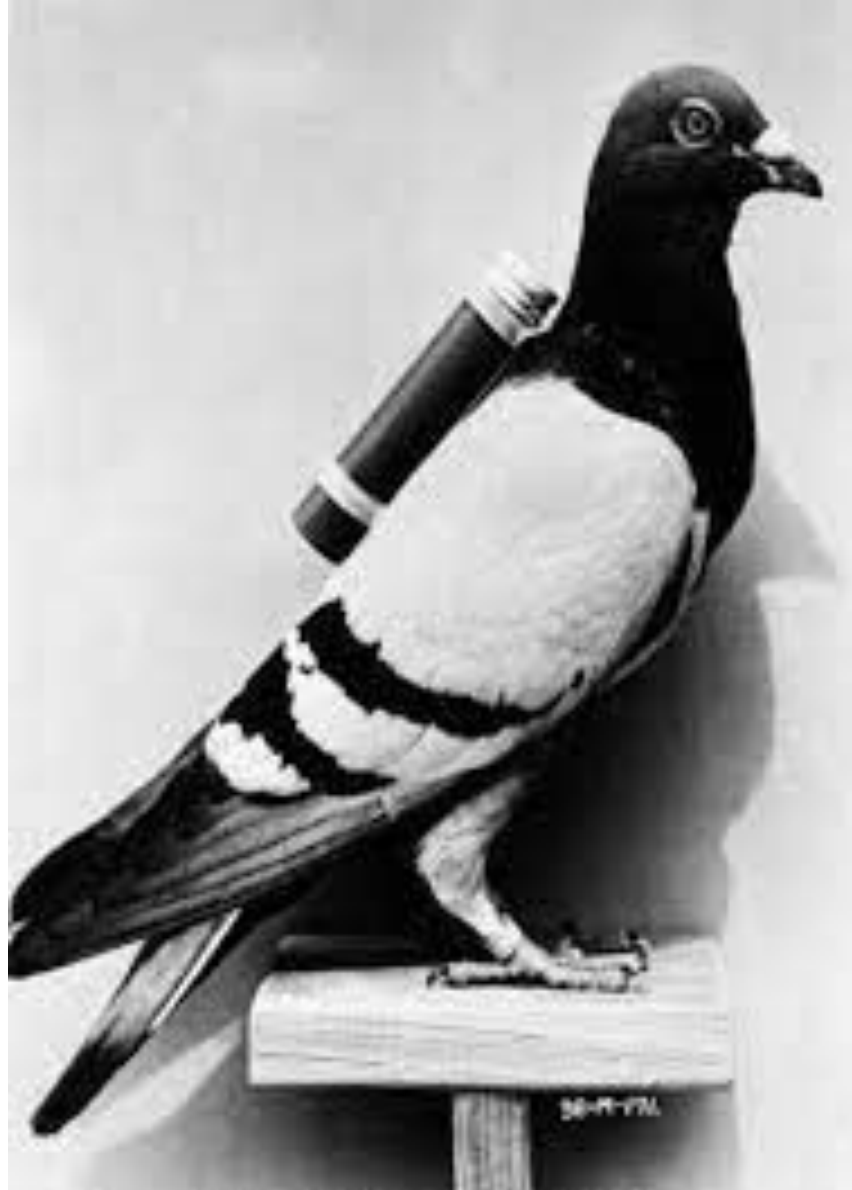
Sinais de fumaça



**Sinalização
por pipas**



Cornetas alpinas



**Pombo-
correio**



Telégrafo



Telex



Rádio

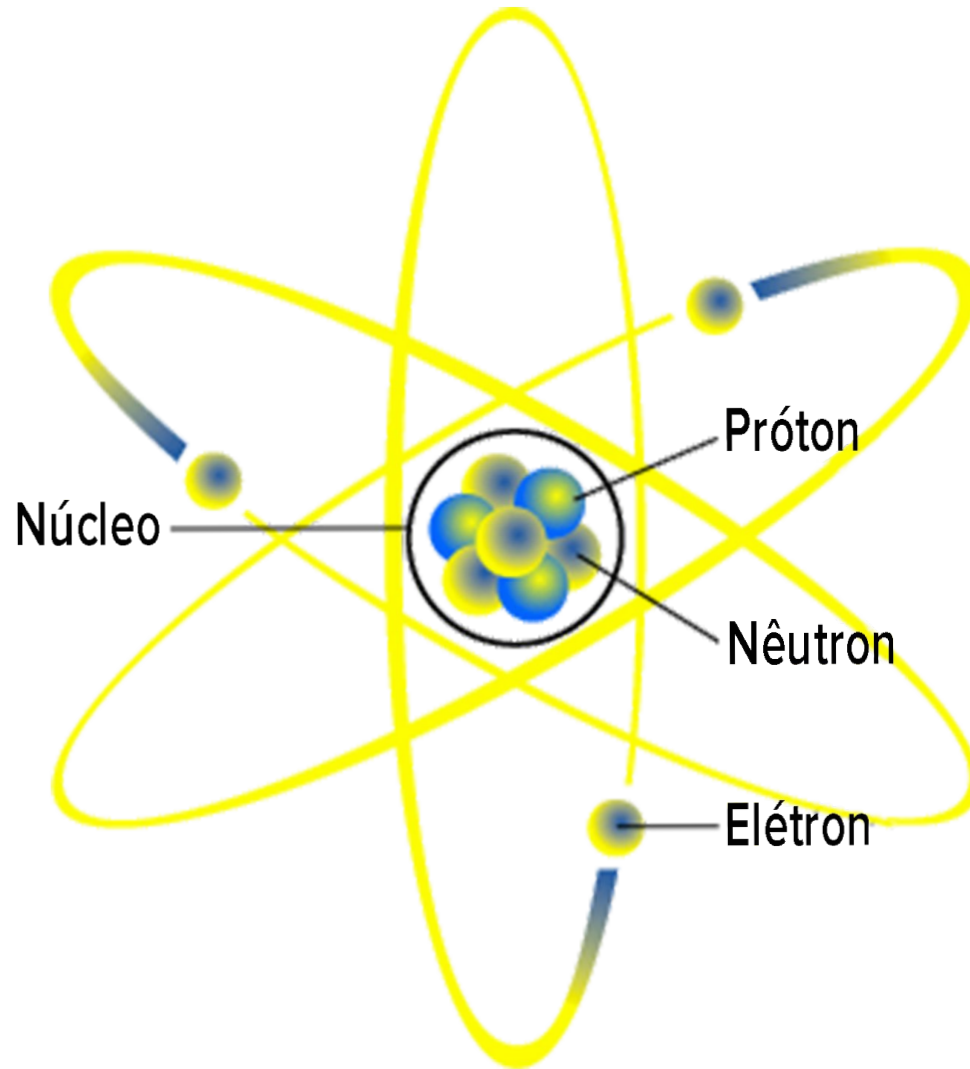


Telephone

Bell's First Telephone

UNDERWOOD & UNDERWOOD, INC.

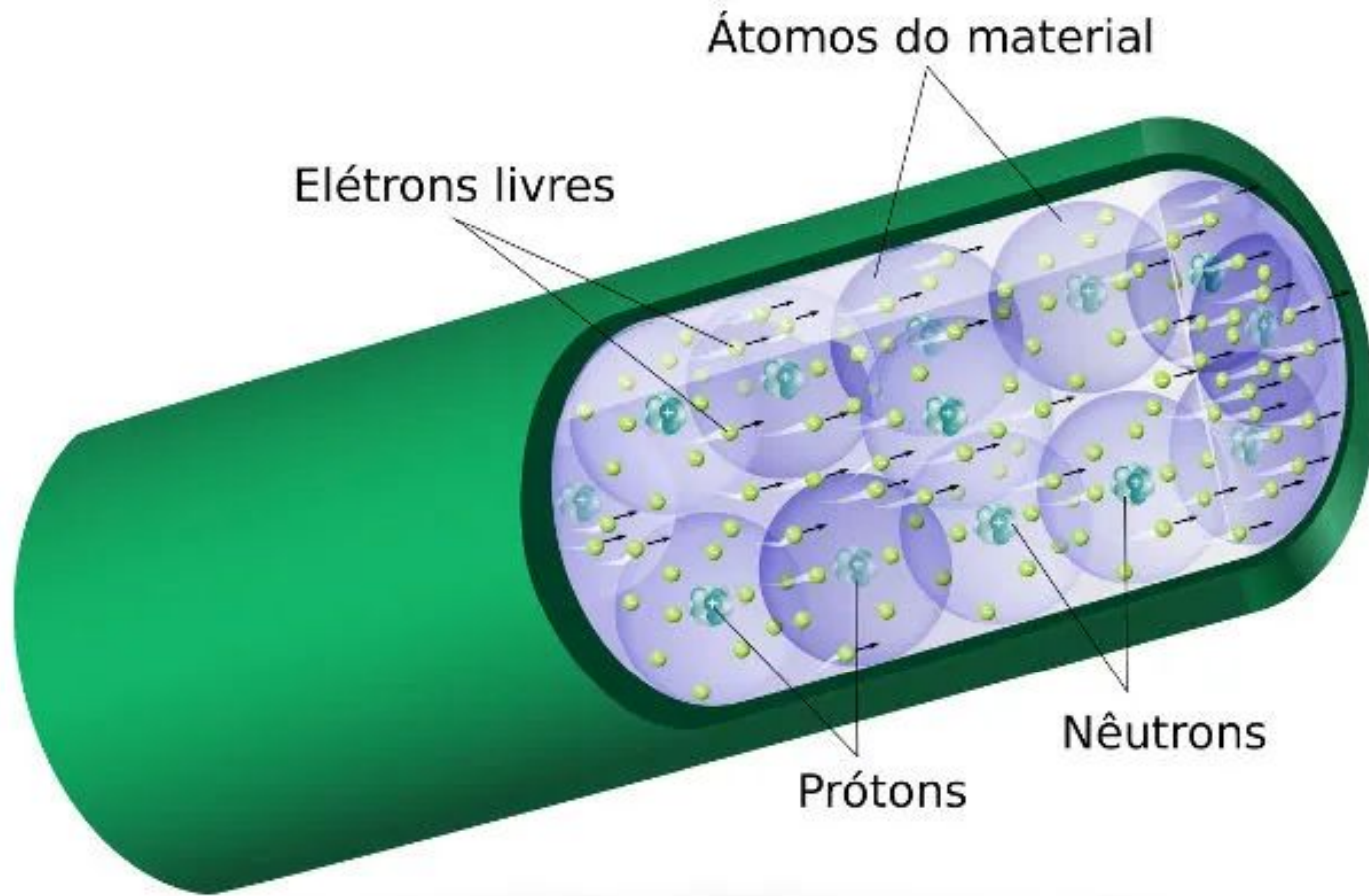
Noções Essenciais de Física para Redes



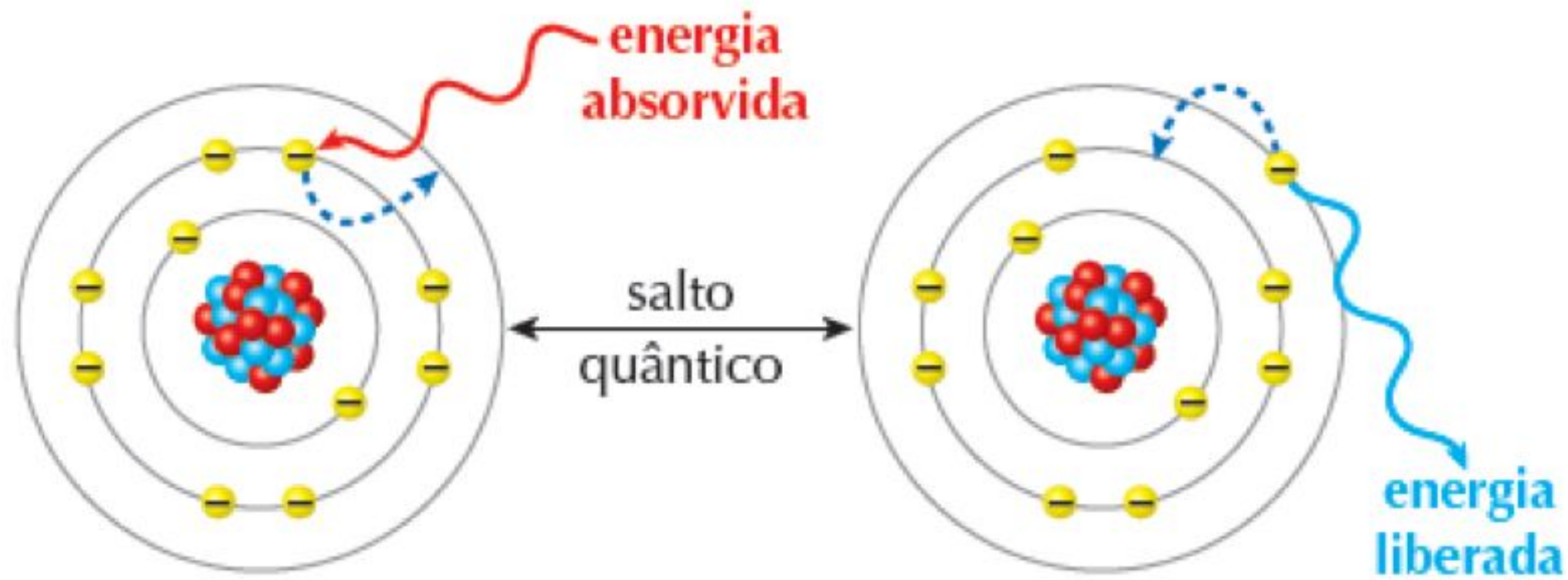
Nêutron: sem carga elétrica

Próton: carga positiva

Elétron: carga negativa



Elétrons são responsáveis pelo transporte de energia elétrica.



**CORPO
ELETRICAMENTE:**

- POSITIVO

$$N_{\text{PRÓTONS}} > N_{\text{ELÉTRONS}}$$



5 p(+)
3 e(-)

- NEUTRO

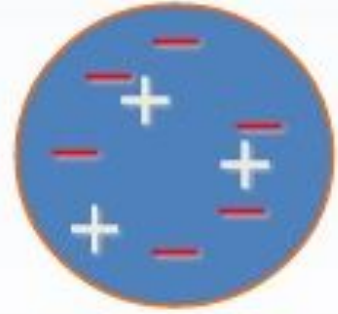
$$N_{\text{PRÓTONS}} = N_{\text{ELÉTRONS}}$$



5 p(+)
5 e(-)

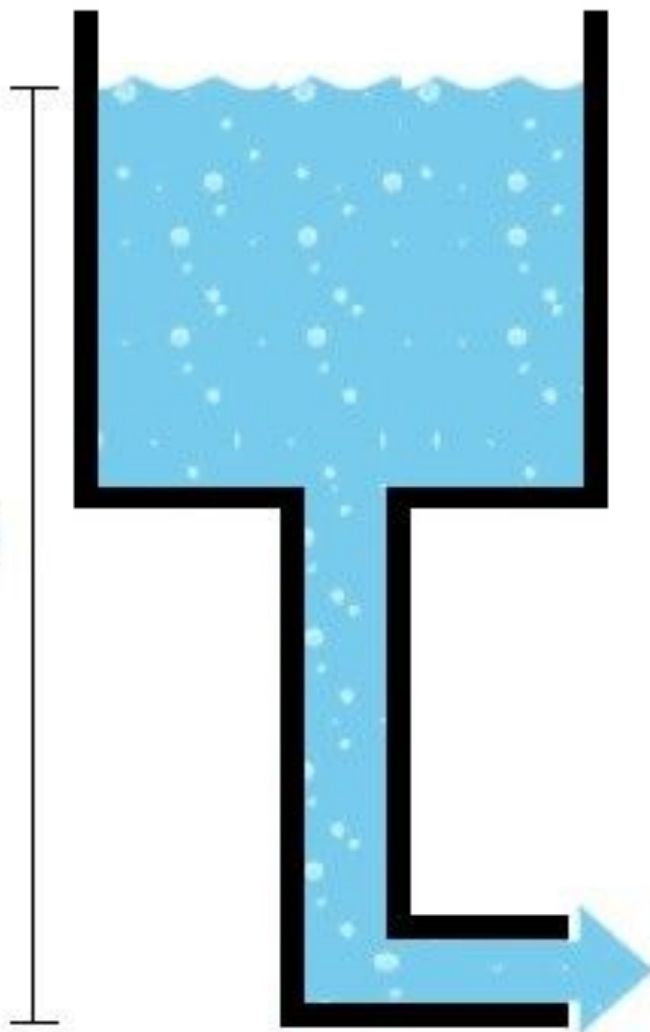
- NEGATIVO

$$N_{\text{PRÓTONS}} < N_{\text{ELÉTRONS}}$$



3 p(+)
6 e(-)

Tensão



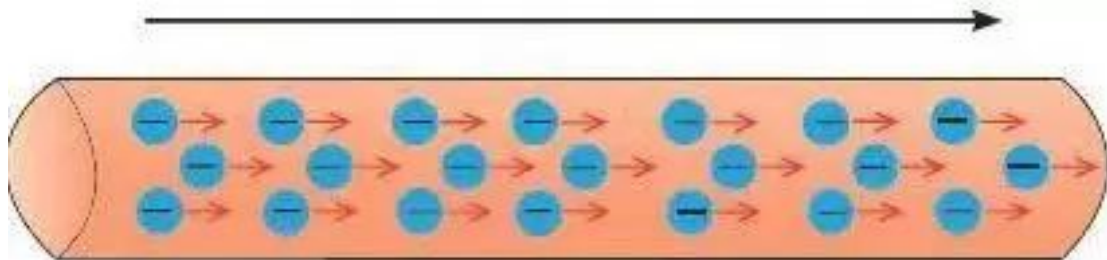
Corrente

Analogia com
Tanque d'água

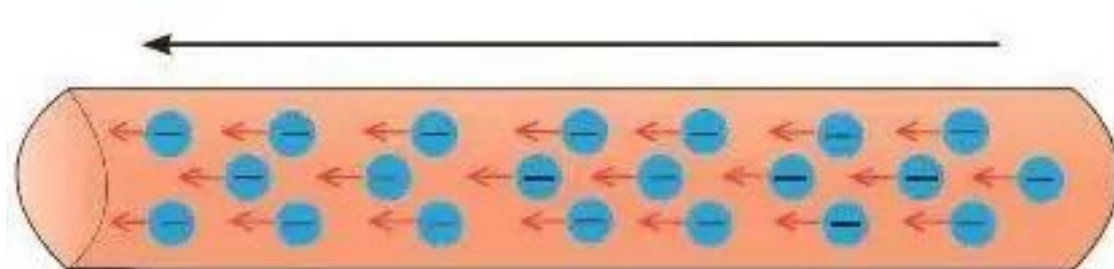
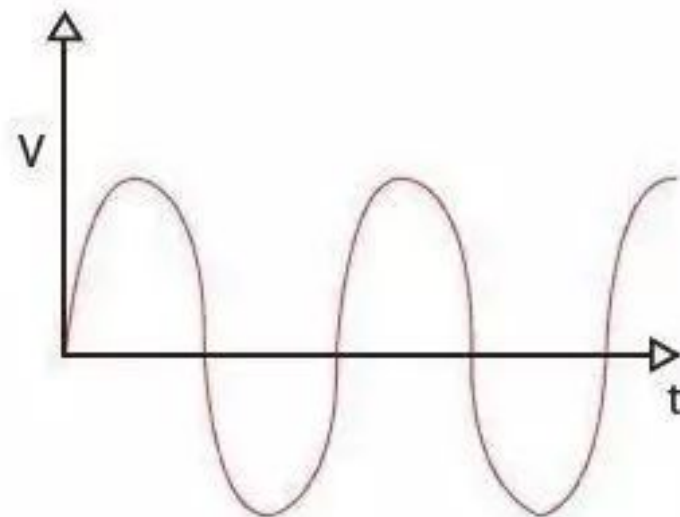
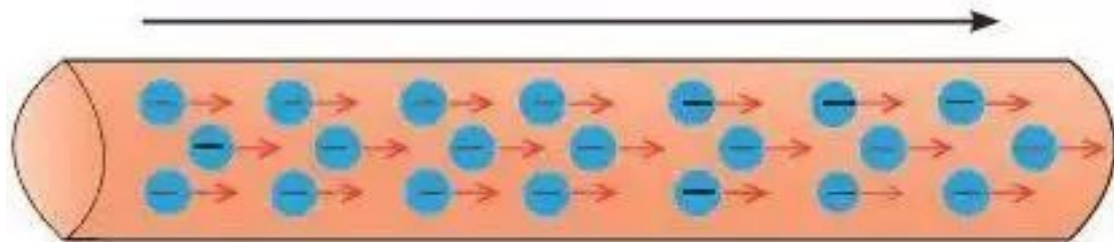


$$R = \frac{\rho \cdot L}{A}$$

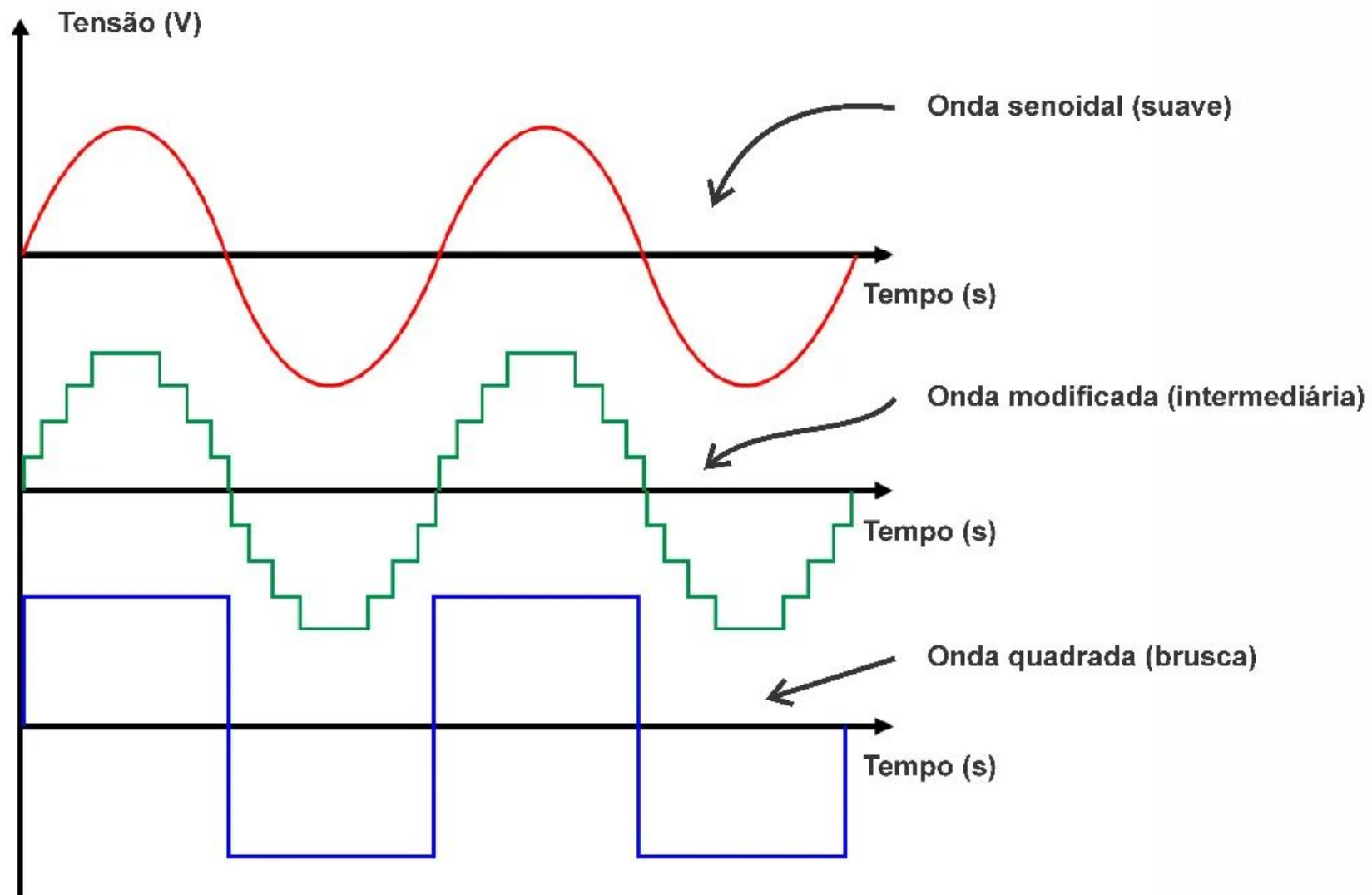
MATERIAL	RESISTIVIDADE ($\Omega \cdot m$)
Prata	$1,6 \times 10^{-8}$
Cobre	$1,7 \times 10^{-8}$
Ouro	$2,4 \times 10^{-8}$
Alumínio	$2,8 \times 10^{-8}$
Chumbo	$2,2 \times 10^{-7}$
Vidro	1×10^{10} a 1×10^{14}
Borracha	$\approx 10^{13}$

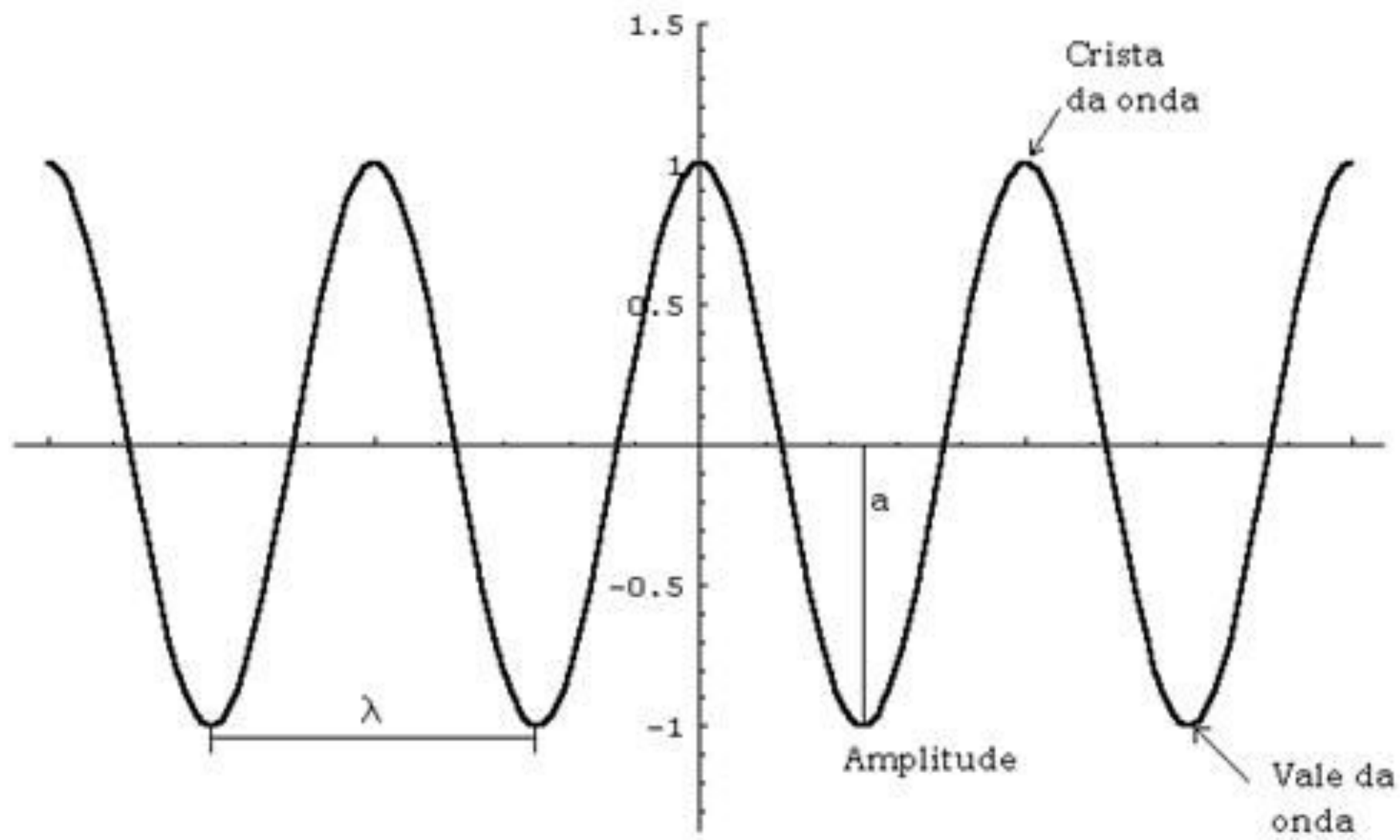


Corrente contínua

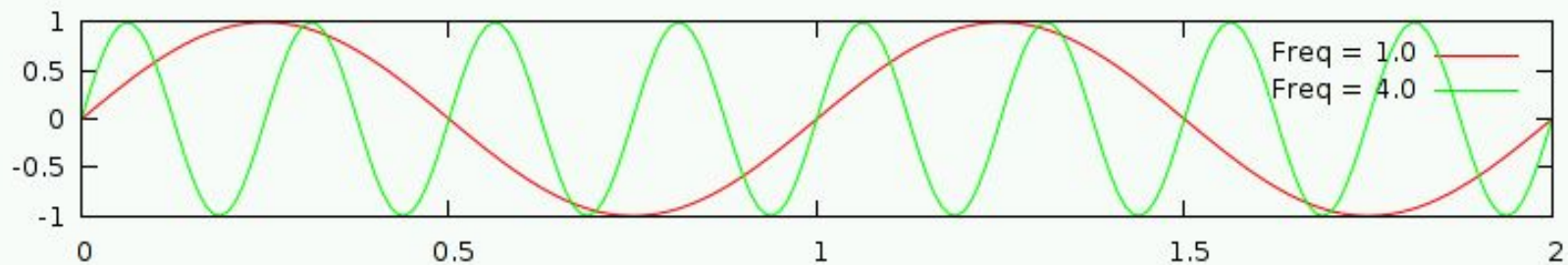


Corrente alternada

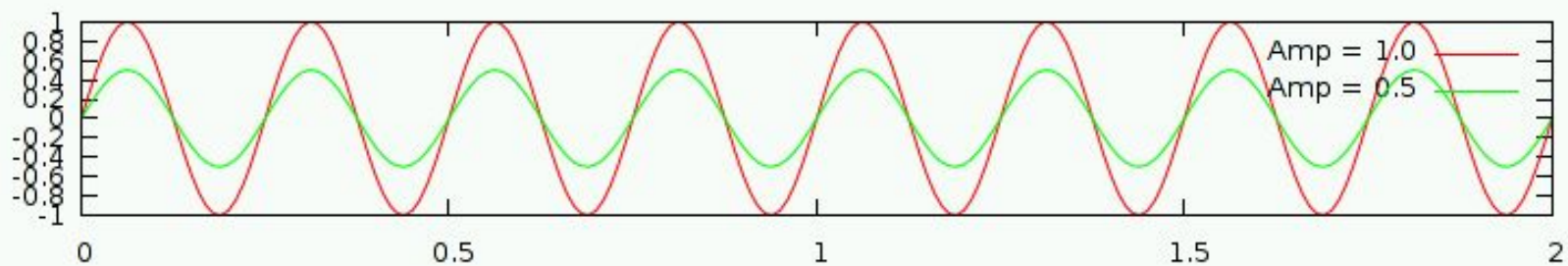




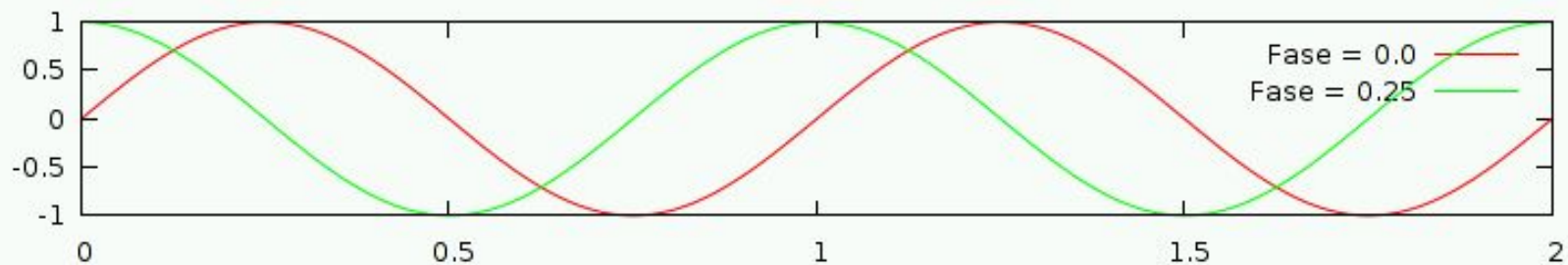
Duas ondas com frequências diferentes



Duas ondas com amplitudes diferentes



Duas ondas com fases diferentes





Ondas de rádio



Micro-ondas



Infravermelho



Luz visível



Ultravioleta



Raios X



Raios gama

